# FLEET TEAM DIGEST

## 737MAX-FTD-29-17003

**Issue Title :** Parker (Abex) Engine-Driven Pump (EDP) V-Band Clamp Band Loop Failures

| Airplane Model | ATA | Minor Model(s) |
|---|---|---|
| 737MAX | 2911-11 | -7, -8, -8200, -9, -BBJ |

| Other Model(s) | ECCN |
|---|---|
| 737NG | 9E991 |

| Originated Date | Last Revised Date | Created On |
|---|---|---|
| 07/03/2017 | 06/15/2022 | 11/29/2021 |

| Estimated Completion Date | Next Update Date |
|---|---|

## Revision Description

Updated the "Status" and "MIlestone" section of the FTD.

## Applicability

All 737MAX aircraft with Parker (Abex) hydraulic pumps.

## Description

In recent months, Boeing has received numerous reports of failed EDP V-band clamps from 737 operators. The vast majority of failures (greater than 95%) have been reported on the P/N 22807 and P/N 24765-1/-01 Parker (Abex) EDP V-Band clamps (Boeing has received a very limited number of failure reports involving P/N 974219 Eaton (Vickers) EDP V-Band clamps). The clamp fractures in fatigue at the location where the sheet metal band is slotted and wrapped around the tee bolt retainer (i.e. the band loop), as shown in the attached photographs. In general, the in-service time associated with these failed clamps is in the range of several hundred flight hours to roughly 10,000 flight hours.

As a result of these failure reports, a decision was made to elevate the project to a Service-Related Problem (SRP), 737NG-SRP-29-0120, to develop appropriate corrective actions to address this issue.

## Background

In order to better understand the reported failures, Boeing conducted an Equipment Quality Analysis (EQA) investigation on several failed in-service clamps. The results of this EQA investigation are as follows:

GENERAL OBSERVATIONS:

==========

- The clamp fractured in the strap that wraps around the T-sleeve

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

- The strap was cut to disassemble the clamp for further investigation. The T-sleeve was manufactured from one piece of sheet which was cut to shape and bent to form the T-sleeve. There were indentations present on the T-sleeve, which appeared to be from contact with tooling during the forming process. It was reported that the T-sleeve appeared to be properly installed on the clamp when it was on the aircraft.

- The T-sleeve was mechanically forced open to examine the interior. There was evidence of bolt thread contact on the interior of the T-sleeve and corresponding damage was found on the bolt threads. Some of the thread tips exhibited flattening.

- The strap fracture initiated on the interior U-shape that divided the strap into two and traveled at an angle to the exterior of the strap. The fracture initiated and propagated by fatigue and the majority of the crack was propagated by fatigue. The final fracture region exhibited evidence of dimple fracture, typical of ultimate separation.

- The strap material was consistent with 301 1/4H stainless steel, as described by AMS 5517. SEM EDS analysis revealed carbon, iron, chromium, nickel, manganese and silicon. The hardness was tested and met the AMS 5517 hardness requirement of 25 HRC min. The strap material was mounted and cross-sectioned to show the microstructure of the material. The microstructure appeared consistent with 301 cold worked stainless steel.

CONCLUSIONS:

==========

1) The clamp fractured in the strap that wraps around the T-sleeve.

2) The fracture initiated and propagated by fatigue until ultimate separation.

3) There was evidence of bolt thread contact on the interior of the T-sleeve.

4) Bolt interference with the T-sleeve may have affected the torque measurement. The required installation torque is printed on the strap, "torque nut to: 70-80 lbs-in."

5) The strap material chemistry, hardness and microstructure were consistent with 301 1/4H, cold worked stainless steel.

As stated above, the metallurgical investigation did not reveal any significant deviations from engineering requirements with regard to the clamp material.

Furthermore, we do not believe that there is a batch-related issue, as we have received failure reports from a wide range of clamp manufacturing batches and there is no correlation to a production date or a quality escapement.

## Status

At this time, Boeing is currently working to make parts and service documents available to the fleet.

Completed Actions:

1. Baseline Engineering Testing (3 Phases) - Completed 11-SEP-2017

2. Root Cause Established - Complete 27-SEP-17

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

** Summary: EDP V-Band Clamp vibration excitation causes trunnion resonance resulting in alternating stresses in the V-Band strap ultimately leading to fatigue failure.

3. Solution Selection - Complete 10-JAN-18

** Summary: Parker developed a prototype riser insert at the T-bolt side band loop. Parker completed initial resonant testing of riser concept and the clamp did not fail. The Boeing and Parker team decided to move forward with additional verification testing of this prototype clamp.

4. Qualification Testing Process

- Current EDP Specification (10-62167) to be updated to include clamp requirements - Completed - DEC-2018

- Test hardware manufacturing - Completed - MAR-2019

- Qualification test procedure development and approval - Completed- MAY-2019

- Qualification testing start - Completed - JUN-2019

- Testing complete and report submitted to Boeing - Completed - SEP-2019

- Service Letter 737-SL-29-163 - Published 2/2/2022

## Interim Action

None at this time.

## Final Action

Boeing service documentation to be released concurrent with production. A new QAD kit is available, P/N 66087AK1, which includes new V-Band Clamp P/N 27763.

## Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

# FLEET TEAM DIGEST

SRP Initiated: 03-APR-2017 (Complete)

Root Cause Established: 29-SEP-2017 (Complete)

Solution Selected: 10-JAN-2018 (Complete)

Change Committed: MAR-2019 (Complete)

Parts Available: FEB-2020 (Complete)

Service Document Service Letter 737-SL-29-163: 2-FEB-2022 (Complete)

## Operator Action

At this time, Boeing will not be recommending fleet-wide inspections of the EDP V-band clamps because in-service experience dictates that a failure of a single EDP V-band clamp band loop is not a safety or operationally-impactful issue. Although we have received many reports of EDP V-band clamp band loop fractures, none of these reports have indicated that the clamp failures resulted in a degradation of EDP performance (i.e. pressure loss), hydraulic fluid leakage, or separation of the pump from the engine gear box. Lastly, every clamp failure reported to Boeing was a result of a mechanic noticing a broken band loop during routine maintenance with the engine cowling opened (i.e. none of the reported occurrences were the result of an in-flight or operations-related failure event). Therefore, Boeing has no objection to operators performing fleet-wide inspections on the subject EDP V-band clamps based on their specific in-service experience; however, a Boeing-recommended inspection will not be imposed at this time.

As stated above, in-service experience suggests that none of the clamp failures resulted in a degradation of EDP performance (i.e. pressure loss), hydraulic fluid leakage, or separation of the pump from the engine gear box. During wind-milling and vibration testing of the CFM56-7B, it was determined that a worst-case clamp failure would result in EDP separation from the engine gear box; however, the EDP would simply remain suspended within the engine cowling via the flexible pressure and supply hoses. In this case, system pressure would remain available via the associated EMDP. 737 hydraulic system architecture ensures that the aircraft will remain fully controllable in the event that an EDP failure (such as separation from the engine gear box) resulting in external hydraulic fluid loss. If the system is breached within the System B EDP circuit, then the system will eventually be lost; however, hydraulic capability will be maintained by System A. Furthermore, if the system is breached within the System A EDP circuit, then the hydraulic fluid reservoir will be drained until the fluid level falls below the system A EDP standpipe. Consequently, fluid flow to the system A EDP circuit will be cut off and further external fluid loss will cease. The system A EMDP will continue to provide system A pressure throughout the remainder of the flight

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

# FLEET TEAM DIGEST

If fractured clamps are found on-aircraft, we recommend the following procedure be carried out when installing a new clamp:

- Attach the EDP with the new V-band clamp:

1) Install the clamp and make sure that it is clear of the engine drain port.

2) Tighten the clamp nut to the value specified on the clamp band.

a. Lightly tap the outer periphery of the V-band clamp with a non-metallic mallet.

NOTE: This will cause the V-band clamp to seat on the pump-gearbox flanges and the nut may loosen.

b. Tighten the nut again to the recommend torque value.

c. Do the steps again until the nut is tightened to the recommended torque value without any movement.

Lastly, operators may choose to procure the new V-Band Clamp from Parker Hannifin Corporation. The new QAD kit available is kit P/N 66087AK1, which includes new V-Band Clamp P/N 27763.

Existing QAD Kit P/N 60827 is two-way interchangeable with P/N 66087AK1. Additionally, existing V-band coupling P/N 24765-01 is two-way interchangeable with P/N 27763.

## Reference Categories

| Reference Type | Reference Number | Revision Number | Reference Date |
|---|---|---|---|
| Fleet Team Economic (FTE) | FTE-29-16013 | | 11/08/2016 |
| Service Related Problem (SRP) | 737NG-SRP-29-0120 | | 04/03/2017 |

## Parts List

| Part Name | Boeing Part Number | Supplier Name | Supplier Number |
|---|---|---|---|
| EDP V-Band Clamp | | Parker | 22807 |
| EDP V-Band Clamp | | Parker | 24765-1/-01 |
| EDP V-Band Clamp | | Parker | 27763 |
| QAD Kit | | Parker | 66087AK1 |

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

# FLEET TEAM DIGEST

## Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- In Service Occurrences : null
- SRPs - Economic : Service Related Problems in work by Boeing that have an economic impact

## Attachments:

a) EDP V-Band Clamp - Typical Failure

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991